

SAMVAAD – TEAM COGNIFY

FULL PRESENTATION SCRIPT – 4 MEMBERS

Simple, natural speaking scripts based on the uploaded 6-slide presentation. Each member has a clearly separated topic with no repeated explanation.

Member 1 – Introduction, Problem & Proposed Solution

Slides 1 & 2

Good morning everyone. We are Team Cognify, and our project is SAMVAAD, an AI-powered Indian Sign Language learning platform.

SAMVAAD is designed to make Indian Sign Language learning more structured, engaging, and practice-oriented. Instead of only watching signs, learners can study a lesson, perform the sign using their device camera, and receive AI-assisted feedback.

The platform provides structured learning through categories, chapters, and step-by-step lessons. It also provides clear visual demonstrations, camera-based practice, AI feedback, and a progress system.

The main idea of SAMVAAD is to close the gap between understanding a sign and confidently performing it.

Now coming to the problem and proposed solution, learners may have limited access to structured ISL learning material. SAMVAAD addresses this through organized categories, chapters, and lessons.

Another problem is passive learning without enough hands-on practice. SAMVAAD provides camera-based practice immediately after learning.

Learners may also find it difficult to know whether a performed sign is correct. Our AI-assisted sign recognition helps by providing a predicted sign and confidence information.

SAMVAAD also provides progress tracking, lesson completion, practice history, XP, streaks, and achievements to encourage consistency.

So, instead of isolated videos or one-time practice, SAMVAAD creates a continuous learning loop for the learner.

Member 2 – Complete User Workflow

Slide 3

Good morning everyone. Now I will explain how a user interacts with SAMVAAD from the beginning to the end.

First, the learner registers or logs into the application and creates their profile. The learner can then personalize their goals, preferences, and language settings.

After that, the learner reaches the home dashboard, where they can see available categories, their learning progress, and reminders.

The learner then selects a category and chooses a lesson. The lesson provides a visual demonstration of the sign so that the learner can understand how it is performed.

After learning the sign, the learner moves directly to the practice stage. They select the sign and open the device camera to perform it.

The system processes the camera input by extracting the required movement landmarks. The AI then analyzes the performed gesture and predicts which sign the learner has performed.

The learner receives the prediction along with confidence information as feedback. Based on this result, the learner can retry the sign or continue to the next activity.

Finally, the attempt and score can be saved, allowing the learner to track their progress. XP, streaks, and achievements also encourage the learner to practice regularly.

So, the complete SAMVAAD journey is: Register, Personalize, Learn, Practice, Analyze, Receive Feedback, and Improve.

Member 3 – Core Modules & Learning Experience

Slide 4

Now I will explain the major modules and features that make SAMVAAD an interactive learning platform.

The first is the Learning Module. It organizes ISL content into categories, chapters, and structured lessons. It also provides visual or animated demonstrations and step-by-step instructions.

The second is the Practice Module. After learning a sign, the learner can select it and practice using the camera. Multiple attempts are supported, and the practice results can be tracked.

The third is the AI Recognition Module. This is one of the important parts of SAMVAAD. MediaPipe is used to extract hand and pose landmarks, while OpenHands is used for ISL gesture recognition. The system then provides the predicted sign and confidence information.

The next module is Feedback and Progress. It helps learners understand their performance and keeps track of lesson completion, practice history, and overall improvement.

SAMVAAD also includes Gamification. XP, daily streaks, achievements, and milestones are used to make learning more engaging and encourage regular practice.

Finally, we have Accessibility features, including large text, high contrast, reduced motion, and personalized preferences.

Together, these modules make SAMVAAD more than just a learning application. They create a complete environment where learners can learn ISL, practice it, understand their performance, and improve continuously.

Member 4 – AI Pipeline, Technology, Innovation & Conclusion

Slides 5 & 6

Finally, I will explain the AI pipeline, system architecture, technology stack, innovation, and expected impact of SAMVAAD.

The AI process starts with the camera, which captures the learner's sign. MediaPipe then extracts hand and pose landmarks from the camera input.

Next, OpenHands recognizes the ISL gesture. The system produces a predicted sign along with confidence information, and this result is shown to the learner as feedback.

For the application architecture, we use Flutter for the frontend. It handles the user interface, navigation, lessons, camera interaction, and 3D or GLB content.

The backend uses Spring Boot for REST APIs, authentication, business logic, and notifications. PostgreSQL is used to store users, lessons, signs, attempts, progress, and achievements.

Therefore, our main technology stack consists of Flutter, Spring Boot, PostgreSQL, MediaPipe, and OpenHands. The overall data flow is: User action, Flutter UI, REST API, Spring Boot service, PostgreSQL, response, and updated progress on the application.

Coming to innovation, SAMVAAD combines learning and immediate practice instead of separating theory from performance. The AI-assisted feedback gives learners information about their performed sign and confidence.

The motivation layer uses XP, streaks, achievements, and progress to encourage consistent learning. The accessible design also supports large text, high contrast, reduced motion, and personalized preferences.

The expected impact of SAMVAAD is more interactive access to ISL learning, immediate practice and performance feedback, support for inclusive communication and ISL awareness, and structured digital learning with measurable progress.

To conclude, SAMVAAD follows a simple cycle: Learn, Practice, Recognize, Receive Feedback, Track Progress, and Improve.

Thank you.